

Module 1

What is 3D Printing? — Tier 1: Beginner

1. What is 3D Printing? (FDM Basics)

3D printing, formally called **additive manufacturing**, builds physical objects by adding material layer by layer — the opposite of traditional manufacturing, which carves material away (subtractive manufacturing, like CNC milling).

The most common and beginner-friendly method is **FDM (Fused Deposition Modeling)**. Here's how it works:

- A spool of plastic filament is fed into the printer
- The filament passes through a heated **hotend**, melting it
- The melted plastic is extruded through a small **nozzle**
- The printer moves the nozzle in precise X/Y/Z paths, depositing plastic in thin layers
- Each layer cools and bonds to the one below it
- Layer by layer, the object takes shape

Why FDM is the standard starting point:

- Lowest cost of entry (printers start under \$200)
- Simplest to operate and maintain
- Widest material selection for beginners
- Huge community and troubleshooting resources

2. Key Terminology

Term	Meaning
Nozzle	The tip that melted plastic is pushed through; sizes typically 0.2–0.8mm
Bed	The flat surface the object is printed on
Filament	The plastic "wire" material, wound on a spool
Layer height	The thickness of each printed layer (affects detail vs. speed)
Extruder	The mechanism that pushes filament into the hotend

Hotend	The heated assembly that melts filament before it exits the nozzle
G-code	The instruction file the printer reads — generated by your slicer (Module 3)

3. Printer Types Overview

FDM vs. Resin (SLA/MSLA)

- **FDM:** uses solid filament, melts and extrudes it — better for functional parts, larger prints, beginner-friendly
- **Resin:** uses liquid resin cured by UV light — much higher detail, better for miniatures/jewelry, but messier and requires post-processing
- **For a true beginner, FDM is the recommended starting point.**

CoreXY vs. Bedslinger

- **Bedslinger:** the bed moves back and forth (Y-axis); print head moves left-right and up-down. Simpler, cheaper, but vibration can lower quality at high speed.
- **CoreXY:** the bed stays still (or only moves vertically); the print head moves on both X and Y via belts. More stable at high speeds, generally pricier.

Enclosed vs. Open Frame

- **Open frame:** no walls around the print area. Cheaper, easier access, more affected by drafts — fine for PLA.
- **Enclosed:** has walls/a chamber. Keeps consistent temperature, essential for warping-prone materials like ABS, reduces noise/fumes.

4. Choosing Your First Printer

Budget Tiers

- **Entry (\$150–300)**: Small build volume, manual or basic auto-leveling, PLA-focused. Great for testing interest.
- **Mid-range (\$300–600)**: Auto bed leveling, larger build volume, often multi-material or enclosure options.
- **Prosumer (\$600+)**: CoreXY, enclosed, multi-material systems (like AMS), faster and more reliable.

Feature Checklist to Look For

- Auto bed leveling (saves huge setup time)
- Community size (bigger community = easier troubleshooting)
- Available spare parts and upgrades
- Enclosure (if you plan to print more than just PLA)
- Multi-material support (if you want multi-color or dissolvable supports later)

Matching to Your Use Case

- **Hobbyist / general learning** → entry to mid-range, open frame is fine
- **Functional parts / engineering prototypes** → mid to prosumer, enclosed preferred
- **Miniatures / high-detail models** → consider resin instead of FDM

5. What to Expect in Your First Week

- **First calibration is mandatory** — bed leveling and first-layer calibration make or break every print after
- **Failed prints are normal** — warping, stringing, and adhesion issues happen to everyone starting out (Module 5 covers troubleshooting)
- **Start small** — print calibration objects and simple test models before anything ambitious
- **Where to get help**: manufacturer community Discord/forum, r/3Dprinting, manufacturer support docs

End of Module 1 — Next: Module 2, Your First Model